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Addendum to My Paper “Algebraic Values of Meromorphic Maps”

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I. Introduction

In my previous paper [1] the following result is crucial (notations are as in [1]):

Existence Theorem. *Let V be a plurisubharmonic function in $\mathbb{C}^n$. There exists $F(z)$ holomorphic in $\mathbb{C}^n$ and not identically zero, such that*

$$\int_{\mathbb{C}^n} |F|^2 e^{-V} (1 + |z|)^{-3n} \omega_n < +\infty.$$

In the proof of this theorem we make use of the fact that if V is plurisubharmonic in an open set Ω then the set of points at which e^{-V} is not summable is closed in Ω and of measure zero. It has been brought to my attention that a proof of this latter assertion does not seem to be available in the literature and it is the purpose of this short note to elucidate this point.

Following the notation of [1] we write

$$\frac{1}{\pi} \partial \bar{\partial} V = T$$

and denote by $\Theta(\|T\|; r, a)$ the average mass of T in the ball $B(r, a)$ of radius r and center a . The function $\Theta(\|T\|; r, a)$ is monotone non-decreasing for $r < \text{dist}(a, \partial\Omega)$; in particular

$$\Theta(\|T\|; a) = \lim_{r \rightarrow 0} \Theta(\|T\|; r, a)$$

exists. $\Theta(\|T\|; a)$ is also called the Lelong number of T at the point a . We note also the inequality

$$\begin{aligned} \left(1 - \frac{|a|}{r}\right)^{2n-2} \Theta(\|T\|; r - |a|, 0) &\leq \Theta(\|T\|; r, a) \\ &\leq \left(1 + \frac{|a|}{r}\right)^{2n-2} \Theta(\|T\|; r + |a|, 0) \end{aligned}$$

(see [1], Lemma 6) which will be used later in the proof of Lemma 1.

We shall prove

Theorem 1. *There exists a constant $\gamma(n) > 0$ depending only on the dimension n , with the property that if*

$$\Theta(\|T\|; a) < \gamma(n)$$

then $e^{-V(z)}$ is summable in some neighborhood of $z = a$.

Note that one has trivially $\Theta(\|T\|; a) = 0$ almost everywhere hence e^{-V} is locally summable outside a set closed in Ω and of measure zero, as was asserted in [1]. Actually much more than this is true and the existence Theorem gives at once:

Theorem 2. *Let V be a plurisubharmonic function in $\mathbb{C}^n$. Then e^{-V} is locally summable outside a complex analytic hypersurface of $\mathbb{C}^n$.*

It is possible to show that e^{-V} is not summable at every point for which

$$\Theta(\|T\|; a) > 2n,$$

by using the argument of [1], (ii) of Lemma 7 together with the fact that one can represent $V(z)$ locally by a Newtonian potential modulo harmonic functions. Combining this result with a local version of our Existence Theorem one gets the following statement of independent interest:

Structure Theorem. *For every positive c the set of points at which*

$$\Theta(\|T\|; a) > c$$

is locally contained in a complex analytic hypersurface. In particular, this set has a locally finite $(2n-2)$ -dimensional Hausdorff measure.

The Structure Theorem and Theorem 1 give immediately a local version of Theorem 2, namely:

Theorem 2A. *Let V be a plurisubharmonic function in an open set Ω . Then e^{-V} is locally summable in $\Omega \setminus N$, where N is closed in Ω and locally contained in a complex analytic hypersurface.*

II. Proof of Theorem 1

Let $T = \frac{1}{\pi} \partial \bar{\partial} V$ and let

$$d\|T\| = iT \wedge \omega_n$$

be the associated positive Radon measure. We shall examine the summability of e^{-V} at the origin; the problem is clearly local so we may assume that V is defined in the ball $|z| < 1$. The Newtonian potential

$$P_\varepsilon(z) = -\frac{(n-2)!}{4\pi^{n-1}} \int_{|\zeta| < \varepsilon} \frac{1}{|z-\zeta|^{2n-2}} d\|T\|(\zeta)$$

differs from $V(z)$ in $|z| < \varepsilon$ only by a harmonic function, so that we shall consider the equivalent problem of the summability of $P_\varepsilon(z)$ at the origin.

Lemma 1. *If $|a| < \varepsilon$ and $r < \varepsilon$ we have*

$$\int_{|z-a|<r} |\nabla P_\varepsilon(z)| \omega_n(z) \leq c_1 \Theta(\|T\|; 3\varepsilon, 0) r^{2n-1}.$$

Proof. We have

$$|\nabla P_\varepsilon(z)| \leq c_2 \int_{|\zeta|<\varepsilon} \frac{1}{|z-\zeta|^{2n-1}} d\|T\|(\zeta)$$

because $d\|T\|$ is a positive measure. Hence

$$\begin{aligned} \int_{|z-a|<r} |\nabla P_\varepsilon(z)| \omega_n(z) &\leq c_2 \int_{|z-a|<r} \int_{|\zeta|<\varepsilon} \frac{1}{|z-\zeta|^{2n-1}} d\|T\|(\zeta) \omega_n(z) \\ &\leq c_2 \int_{|z-a|<r} \int_{|\zeta-a|<2\varepsilon} \frac{1}{|z-\zeta|^{2n-1}} d\|T\|(\zeta) \omega_n(z) \\ &= c_2 I. \end{aligned}$$

Next, one checks easily that

$$\int_{|z-a|<r} \frac{\omega_n(z)}{|z-\zeta|^{2n-1}} \leq c_3 r(t+1)^{-2n+1}$$

where

$$t = \frac{1}{r} |\zeta - a|;$$

for if $t \geq 2$ then $|z - \zeta| \geq r(t - 1)$, while if $t < 2$ then $|z - a| < r$ is contained in $|z - \zeta| < 3r$ and the computation again becomes obvious. We deduce the bound

$$\begin{aligned} I &\leq c_3 r \int_{|\zeta-a|<2\varepsilon} \left(1 + \frac{1}{r} |\zeta - a|\right)^{-2n+1} d\|T\|(\zeta) \\ &= c_3 r \int_0^{2\varepsilon/r} (1+t)^{-2n+1} d\|T\| B(rt, a) \\ &= \frac{c_3 r}{\left(1 + \frac{2\varepsilon}{r}\right)^{2n-1}} \|T\| B(2\varepsilon, a) + c_3 r(2n-1) \int_0^{2\varepsilon/r} \frac{\|T\| B(rt, a)}{(1+t)^{2n}} dt. \end{aligned}$$

Finally

$$\begin{aligned} \|T\| B(rt, a) &= \frac{\pi^{n-1}}{(n-1)!} (rt)^{2n-2} \Theta(\|T\|; rt, a) \\ &\leq \frac{\pi^{n-1}}{(n-1)!} (rt)^{2n-2} \Theta(\|T\|; 2\varepsilon, a) \\ &\leq c_4 (rt)^{2n-2} \Theta(\|T\|; 3\varepsilon, 0) \end{aligned}$$

the last inequality being one noted earlier. Combining this estimate with the bound already obtained for I , Lemma 1 follows.

Lemma 2 (John and Nirenberg). *Let $U(z)$ be a summable function in $|z| < 2R$, such that*

$$\int_{|z-a|<r} |\nabla U(z)| \omega_n(z) \leq \gamma r^{2n-1}$$

for every a, r satisfying $|a| < R, r < R$.

Then we have

$$\min_k \int_{|z|<\frac{1}{2}R} \exp(c_5 \gamma^{-1} |U(z) - k|) \omega_n(z) \leq c_6 R^{2n},$$

where c_5, c_6 depend only on n .

Proof. This is a special case of a famous result of John and Nirenberg [2]; a simple proof of Lemma 2 has been found recently by Trudinger [3].

Combining Lemmas 1 and 2 we get that $\exp(-c_5 \gamma^{-1} P_\varepsilon(z))$, and hence $\exp(-c_5 \gamma^{-1} V(z))$, is summable in $|z| < \varepsilon/2$ with γ given by

$$\gamma = c_1 \Theta(\|T\|; 3\varepsilon, 0).$$

Finally assume $\Theta(\|T\|; 0) < c_5/c_1 = \gamma(n)$. Then there exists $\varepsilon > 0$ such that $\Theta(\|T\|; 3\varepsilon, 0) < c_5/c_1$ and $\exp(-V(z))$ will be summable in a neighborhood of the origin; this completes the proof of Theorem 1.

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